

Task: Credit Card Churn Analysis

Initial Data Assessment & Insights:

- **Target Imbalance:** The target variable (Exited) is imbalanced. About **79.6%** of customers stayed, while **20.4%** churned.
- **Geography:** Customers from Germany have a significantly higher churn rate (~32.4%) compared to France and Spain (~16%).
- **Gender:** Female customers are churning at a higher rate (~25%) than male customers (~16%).
- **Age:** Older customers (specifically in the 40–60 age bracket) show a much higher likelihood of churning.

Data Preprocessing

Before feeding this data into any model, we must preprocess it:

1. **Dropping Useless Columns:** RowNumber, CustomerId, and Surname do not provide predictive value.
2. **Encode Categorical Variables:** Using One-Hot Encoding (pd.get_dummies) for Geography and Gender.
3. **Feature Scaling:** Standardizing our numerical columns (like Balance, EstimatedSalary, Age) using StandardScaler. This is strictly required for Distance-based models, Neural Networks, and Logistic Regression.
4. **Handle Class Imbalance:** Since churners are only 20% of the dataset, considering using **SMOTE** (Synthetic Minority Over-sampling Technique) to balance our training data, or utilize class_weight='balanced' inside our model algorithms.

Step B: Machine Learning Models (Recommended)

Tree-based ensemble models usually perform best on this type of tabular data with minimal tuning.

- **Random Forest Classifier:** A great baseline. It's robust to outliers, doesn't strictly require feature scaling, and allows you to easily extract

"Feature Importances" to see which variables drive churn the most (e.g., Age and Balance).

- **XGBoost or LightGBM:** These gradient boosting algorithms will likely yield the highest accuracy and F1-score for this dataset.